

Reversing Lamps

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)	
 ST3093-A	Fluke 77-1V Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

When the transmission is in reverse, the PCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the transmission is in reverse. The BCM provides voltage to the reversing lamps when it receives the message that the transmission is in reverse and the ignition is in RUN.

Field-Effect Transistor (FET) Protection

A Field-Effect Transistor (FET) is a type of transistor that, when used with module software, monitors and controls current flow on module outputs. The FET protection strategy prevents module damage in the event of excessive current flow.

The BCM utilizes a FET protective circuit strategy for many of its outputs (such as a headlamp output circuit). Output loads (current level) are monitored for excessive current (typically short circuits) and are shut down when a fault event is detected. A short circuit DTC is stored at the fault event and a cumulative counter is started.

When the demand for the output is no longer present, the module resets the FET protection, allowing the circuit to function. If the circuit is still shorted the next time the driver requests a circuit to activate that has been shut down by a previous short (FET protection), the FET protection shuts off the circuit again and the cumulative counter advances.

When the excessive circuit load occurs often enough, the module shuts down the output until a repair procedure is carried out. Each FET protected circuit has 3 predefined levels of short circuit tolerance based on the harmful effect of each circuit fault on the FET and the ability of the FET to withstand it. A module lifetime level of fault events is established based upon the durability of the FET . If the total tolerance level is determined to be 600 fault events, the 3 predefined levels would be 200, 400 and 600 fault events.

When a tolerance level is reached, the short circuit DTC that was stored on the first failure cannot be cleared by the clear the continuous DTCs command. The module does not allow this code to be cleared or the circuit restored to normal operation until a successful self-test proves that the fault has been repaired. After the self-test has successfully completed (no on-

demand DTCs present), DTC U1000:00 and the associated DTC (the DTC related to the shorted circuit) automatically clears and the circuit function returns. The module never resets the fault event counter to zero and continues to advance the fault event counter as short circuit fault events occur.

If the number of short circuit fault events reach the third level, then DTCs U1000:00 and U3000:49 set along with the associated short circuit DTC. DTC U3000:49 cannot be cleared and the module must be replaced after the repair.

The BCM FET protected output circuits for the reverse lamp system is the reverse lamps output circuit.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of electrical damage.

Visual Inspection Chart

Electrical
• Bulb • Wiring, terminals or connectors

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:
 - verify the ignition is on.
 - The air bag warning indicator prove-out confirms ignition on (other indicators may not prove ignition on).
If the ignition does not turn on, refer to Section 211-05 to diagnose no power in run.
 - verify the scan tool operation with a known good vehicle.
 - refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
 - If the network test passes, retrieve and record the continuous memory DTCs.

8. Clear the continuous DTCs and carry out the self-test diagnostics for the BCM and the PCM.

9. If the DTCs retrieved are related to the concern, refer to Diagnostic Trouble Code (DTC) Chart in this section. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
Both reversing lamps are inoperative	- Wiring, terminals or connectors - Transmission gear input - Body Control Module (BCM)	VERIFY the bulbs are OK. If OK, GO to Pinpoint Test U.
An individual reversing lamp is inoperative	- Wiring, terminals or connectors - Bulb socket	VERIFY the bulb is OK. If OK, GO to Pinpoint Test V.
The reversing lamps are on continuously	- Wiring, terminals or connectors - Auto-dimming interior mirror - Transmission gear input - BCM	GO to Pinpoint Test W.

Pinpoint Tests

Pinpoint Test U: Both Reversing Lamps Are Inoperative

Refer to Wiring Diagrams Cell [93](#) , Reversing Lamps for schematic and connector information.

Normal Operation

The PCM monitors the transmission gear status. When the transmission is in reverse, the PCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the transmission is in reverse. When the BCM receives the message indicating the transmission is in reverse and the ignition is in RUN, the BCM provides voltage to the reversing lamps.

The reversing lamps share the fused power supply with the high mounted stoplamp and the ground circuit with the rear stoplamps.

- DTC B1277:11 (Reverse Lamp: Circuit Short To Ground) — This DTC sets when the BCM detects a short to ground from the reversing lamp output circuit.
- DTC B1277:15 (Reverse Lamp: Circuit Short To Battery or Open) — This DTC sets when the BCM detects an open from the reversing lamp output circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Transmission gear input
- BCM

PINPOINT TEST U : BOTH REVERSING LAMPS ARE INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

U1 CHECK THE STOPLAMP OPERATION

- Ignition OFF.
- Apply the brake pedal while observing all the stoplamps.

Do the stoplamps illuminate?

Yes	GO to U2 .
No	GO to Pinpoint Test I.

U2 CHECK THE BCM OUTPUT

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: BCM DataLogger .
- Select the BCM reverse lamps active command (RVRS_LMPS). Command the reverse lamps on then off.

Do the reversing lamps illuminate when commanded on?

Yes	REFER to Section 307-01 to diagnose a Transmission Range (TR) concern.
No	GO to U3 .

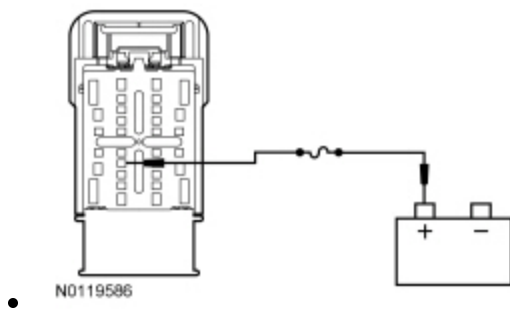
U3 BYPASS THE BCM

NOTE: If the jumper wire fails, refer to the Wiring Diagrams manual to identify the possible causes of the circuit short. After the repair:

*If no DTCs are present, test the system for normal operation. If DTC U1000:00 is present, clear the DTCs **and** repeat the self-test (**required to enable the reversing lamp output driver if DTC U1000:00 is present**). Test the system for normal operation.*

If DTC U3000:49 is present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . Test the system for normal operation.

- Ignition OFF.
- Disconnect: BCM [C2280D](#) .
- Connect a fused jumper wire between the BCM [C2280D](#) Pin 13, circuit CLS28 (BU/WH), harness side and battery positive.



Do the reversing lamps illuminate?

Yes	REMOVE the jumper wire. GO to U4 .
No	REMOVE the jumper wire. REPAIR circuit CLS28 (BU/WH) for an open. TEST the system for normal operation.

U4 CHECK FOR CORRECT BCM OPERATION

- Disconnect the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test V: An Individual Reversing Lamp Is Inoperative

Refer to Wiring Diagrams Cell [93](#) , Reversing Lamps for schematic and connector information.

Normal Operation

When the transmission is in reverse, the PCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the transmission is in reverse. When the BCM receives the message indicating the transmission is in reverse, the BCM provides voltage to the reversing lamps.

This pinpoint test is intended to diagnose the following:

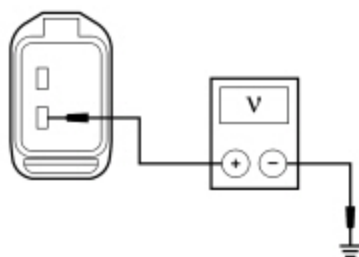
- Wiring, terminals or connectors
- Bulb socket

PINPOINT TEST V : AN INDIVIDUAL REVERSING LAMP IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

V1 CHECK FOR VOLTAGE TO THE REVERSING LAMP

- Ignition OFF.
- Apply the parking brake.
- Disconnect: Inoperative Reversing Lamp .
- Ignition ON.
- Select REVERSE.
- Measure the voltage between the LH reversing lamp [C487](#) Pin 1, circuit CLS28 (BU/WH), harness side and ground; or between the RH reversing lamp [C488](#) Pin 1, circuit CLS28 (BU/WH), harness side and ground.



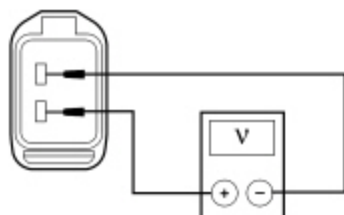
• A0057554

Is the voltage greater than 10 volts?

Yes	GO to V2 .
No	REPAIR circuit CLS28 (BU/WH) for an open. TEST the system for normal operation.

V2 CHECK THE REVERSING LAMP GROUND CIRCUIT FOR AN OPEN

- Measure the voltage between the LH reversing lamp [C487](#) Pin 1, circuit CLS28 (BU/WH), harness side and the LH reversing lamp [C487](#) Pin 2, circuit GD117 (BK), harness side; or between the RH reversing lamp [C488](#) Pin 1, circuit CLS28 (BU/WH), harness side and the RH reversing lamp [C488](#) Pin 2, circuit GD117 (BK), harness side.



• N0088182

Is the voltage greater than 10 volts?

Yes	INSTALL a new bulb socket. TEST the system for normal operation.
No	REPAIR circuit GD117 (BK) for an open. TEST the system for normal operation.

Pinpoint Test W: The Reversing Lamps Are On Continuously

Refer to Wiring Diagrams Cell [93](#) , Reversing Lamps for schematic and connector information.

Normal Operation

The PCM monitors the transmission gear status. When the transmission is in reverse, the PCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) , indicating the transmission is in reverse. When the BCM receives the message indicating the transmission is in reverse and the ignition is in RUN, the BCM provides voltage to the reversing lamps and the auto-dimming interior mirror.

- DTC B1277:15 (Reverse Lamp: Circuit Short To Battery or Open) — This DTC sets when the BCM detects a short to voltage from the reversing lamp output circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Auto-dimming interior mirror
- Transmission gear input
- BCM

PINPOINT TEST W : THE REVERSING LAMPS ARE ON CONTINUOUSLY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
W1 CHECK THE BCM CONTROL OF THE REVERSING LAMPS	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: BCM DataLogger . • Select the BCM reverse lamps active command (RVRS_LMPS). Command the reverse lamps off.	
Do the reversing lamps turn off?	
Yes	REFER to Section 307-01 to diagnose a Transmission Range (TR) concern.
No	GO to W2 .
W2 CHECK THE BCM OUTPUT	
• Ignition OFF. • Disconnect: BCM C2280D . • Ignition ON.	
Do the reversing lamps continue to illuminate?	
Yes	REPAIR circuit CLS28 (BU/WH) for a short to voltage. TEST the system for normal operation.
No	GO to W3 .

W3 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.